

Derivation v29: The β Control Parameter

BLOCK-003 Gravity Program — $\beta = \sigma L^2 / \bar{M}_{\text{Pl}}^2$

EDC Collaboration

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Abstract

We derive and evaluate the dimensionless control parameter $\beta \equiv \sigma L^2 / \bar{M}_{\text{Pl}}^2$ that governs the KK mass gap in the brane-world scenario. Building on v26–v28, where $b = \lambda\beta$ controls the Robin boundary condition spectrum via $\tan(x) = -b/x$, we provide two independent derivation routes for β . Route A computes β directly from definitions; Route B infers β from the spectral equation given λ . We address all convention issues (Planck mass, interval length, SI vs natural units) and provide a complete uncertainty budget. All reviewer traps (TRAP-1 through TRAP-10) are explicitly addressed with equation references. The gap remains **[I]**+**[BL]** until internal derivation of L is achieved. Numerical verification confirms all formulas.

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1 Conventions and Units

1.1 Unit System Declaration

Throughout this note, we work in **natural units** where:

$$\hbar = c = 1 \quad (1)$$

In these units:

$$[\text{length}] = [\text{time}] = [\text{energy}]^{-1} = [\text{mass}]^{-1} \quad (2)$$

$$[\text{mass}] = [\text{energy}] = [\text{momentum}] \quad (3)$$

When SI units are needed (for metrology), we restore $\hbar$ and c explicitly.

1.2 Planck Mass Conventions

Two conventions exist in the literature:

Convention 1.1 (Reduced Planck Mass).

$$\bar{M}_{\text{Pl}} \equiv \sqrt{\frac{\hbar c}{8\pi G_N}} = 2.435 \times 10^{18} \text{ GeV} \quad (4)$$

Convention 1.2 (Original Planck Mass).

$$M_{\text{Pl}} \equiv \sqrt{\frac{\hbar c}{G_N}} = 1.221 \times 10^{19} \text{ GeV} \quad (5)$$

The relation between them:

$$\boxed{M_{\text{Pl}} = \sqrt{8\pi} \bar{M}_{\text{Pl}} \approx 5.013 \bar{M}_{\text{Pl}}} \quad (6)$$

1.3 5D Planck Mass Convention Map

From the bridge relation $\bar{M}_{\text{Pl}}^2 = M_5^3 L$, the 5D Planck mass inherits the convention:

$$M_5^{(\text{red})} = \left(\frac{\bar{M}_{\text{Pl}}^2}{L} \right)^{1/3}, \quad M_5^{(\text{orig})} = \left(\frac{M_{\text{Pl}}^2}{L} \right)^{1/3} \quad (7)$$

The map between them:

$$\boxed{M_5^{(\text{orig})} = (8\pi)^{1/3} M_5^{(\text{red})} \approx 2.929 M_5^{(\text{red})}} \quad (8)$$

Convention adopted: We use $\bar{M}_{\text{Pl}}$ (reduced) throughout, consistent with v20–v24.

1.4 Length Conventions: L vs R vs R_ξ

Convention 1.3 (Canonical Length Dictionary (v22+)).

$$L \equiv \text{interval length on } [0, L] \quad (\text{canonical}) \quad (9)$$

$$R \equiv \frac{L}{\pi} \quad (\text{old circle radius convention}) \quad (10)$$

$$R_\xi \equiv L \quad (\text{canonical EDC notation}) \quad (11)$$

The π -factor relates the conventions:

$$L = \pi R, \quad R_\xi = L = \pi R \quad (12)$$

Physical interpretation: L is the proper length of the extra dimension interval.

1.5 Spectral Equation Conventions

The transcendental eigenvalue equation from v26:

$$\tan(x_n) = -\frac{b}{x_n}, \quad x_n \equiv m_n L \quad (13)$$

With $b = m_b L$ and the Robin boundary condition $\psi'(L) = m_b \psi(L)$.

1.6 Brane Tension Dimensions

In natural units ($\hbar = c = 1$):

$$[\sigma] = M^4 = (\text{energy})^4 = (\text{length})^{-4} \quad (14)$$

In SI units:

$$[\sigma]_{\text{SI}} = \frac{\text{energy}}{\text{volume}} = \frac{\text{kg}}{\text{m} \cdot \text{s}^2} \quad (15)$$

The conversion:

$$\sigma_{\text{SI}} = \sigma_{\text{natural}} \cdot \frac{(\hbar c)^{-3}}{1} \cdot \frac{1}{c^4} \cdot \hbar c = \sigma_{\text{natural}} \cdot \frac{1}{(\hbar c)^3 / (\hbar c)} = \sigma_{\text{natural}} \cdot (\hbar c)^{-3} \quad (16)$$

2 Definition of β

2.1 Primary Definition

Definition 2.1 (Control Parameter β).

$$\boxed{\beta \equiv \frac{\sigma L^2}{\bar{M}_{\text{Pl}}^2}} \quad (17)$$

2.2 Dimensional Verification (Natural Units)

Check dimensions in $\hbar = c = 1$:

$$[\beta] = \frac{[\sigma] \cdot [L]^2}{[\bar{M}_{\text{Pl}}]^2} = \frac{M^4 \cdot M^{-2}}{M^2} = \frac{M^2}{M^2} = 1 \quad \checkmark \quad (18)$$

β is **dimensionless** as required for a control parameter.

2.3 Connection to v27: $b = \lambda\beta$

From v27, the Robin parameter is:

$$b = m_b L = \lambda \frac{\sigma}{M_5^3} \cdot L = \lambda \frac{\sigma L}{M_5^3} \quad (19)$$

Using the bridge relation $\bar{M}_{\text{Pl}}^2 = M_5^3 L$:

$$M_5^3 = \frac{\bar{M}_{\text{Pl}}^2}{L} \quad (20)$$

Substituting:

$$b = \lambda \frac{\sigma L}{\bar{M}_{\text{Pl}}^2 / L} = \lambda \frac{\sigma L^2}{\bar{M}_{\text{Pl}}^2} = \lambda\beta \quad (21)$$

$$\boxed{b = \lambda\beta} \quad (22)$$

This confirms consistency with the v27 derivation.

2.4 Physical Interpretation of β

β measures the *brane tension relative to 4D gravity at the compactification scale*:

$$\beta = \frac{\sigma L^2}{\bar{M}_{\text{Pl}}^2} = \frac{(\text{brane energy density}) \times (\text{area})}{(4\text{D gravity scale})^2} \quad (23)$$

Alternative interpretation using M_5 :

$$\beta = \frac{\sigma L}{M_5^3} = \frac{\sigma L}{\bar{M}_{\text{Pl}}^2/L} \cdot \frac{L}{\bar{M}_{\text{Pl}}^2} \cdot \bar{M}_{\text{Pl}}^2 = \frac{\sigma}{M_5^3} \cdot L \quad (24)$$

3 Route A: Direct Derivation of β

3.1 Starting Point

We derive β directly from its definition using the metrological anchor.

3.2 Metrological Anchor: $\sigma L^3 = \hbar c$

From EDC foundational relations (Part I), the brane tension and compactification scale are connected to Planck's constant:

$$\sigma L^3 = \hbar c \quad [\text{BL}] \quad (25)$$

Tag justification: This uses the SI definition of $\hbar$ (exact since 2019), but L is not internally derived, so the relation is **[BL]**.

3.3 Solving for σ

From eq. (25):

$$\sigma = \frac{\hbar c}{L^3} \quad (26)$$

3.4 Substituting into β

$$\beta = \frac{\sigma L^2}{\bar{M}_{\text{Pl}}^2} = \frac{1}{\bar{M}_{\text{Pl}}^2} \cdot \frac{\hbar c}{L^3} \cdot L^2 = \frac{\hbar c}{L \bar{M}_{\text{Pl}}^2} \quad (27)$$

$$\boxed{\beta = \frac{\hbar c}{L \bar{M}_{\text{Pl}}^2} \quad [\text{BL}]} \quad (28)$$

3.5 Dimensional Check (SI Units)

In SI:

$$[\hbar c] = \text{J} \cdot \text{s} \cdot \frac{\text{m}}{\text{s}} = \text{J} \cdot \text{m} \quad (29)$$

$$[L] = \text{m} \quad (30)$$

$$[\bar{M}_{\text{Pl}}^2] = (\text{energy})^2 = \text{J}^2 \quad (\text{in energy units}) \quad (31)$$

Converting $\bar{M}_{\text{Pl}}$ to SI energy:

$$\bar{M}_{\text{Pl}} = 2.435 \times 10^{18} \text{ GeV} = 2.435 \times 10^{18} \times 1.602 \times 10^{-10} \text{ J} = 3.90 \times 10^8 \text{ J} \quad (32)$$

Then:

$$[\beta]_{\text{SI}} = \frac{\text{J} \cdot \text{m}}{\text{m} \cdot \text{J}^2} = \frac{1}{\text{J}} \neq 1 \quad (33)$$

Resolution: The formula $\beta = \hbar c / (L \bar{M}_{\text{Pl}}^2)$ is valid in natural units. In SI, we must use consistent energy-length units:

$$\beta = \frac{\hbar c}{L \cdot (\bar{M}_{\text{Pl}} c^2)^2 / c^4} = \frac{\hbar c^5}{L \cdot (\bar{M}_{\text{Pl}} c^2)^2} \quad (34)$$

Or equivalently, work entirely in natural units where $\hbar = c = 1$.

3.6 Numeric Evaluation with L Open

Leaving L as a free scale, define:

$$\bar{M}_{\text{Pl}}^2 = (2.435 \times 10^{18} \text{ GeV})^2 = 5.929 \times 10^{36} \text{ GeV}^2 \quad (35)$$

In SI units:

$$\hbar c = 1.055 \times 10^{-34} \text{ J} \cdot \text{s} \times 3 \times 10^8 \text{ m/s} = 3.16 \times 10^{-26} \text{ J} \cdot \text{m} \quad (36)$$

Combining:

$$\beta(L) = \frac{\hbar c}{L \bar{M}_{\text{Pl}}^2} = \frac{3.16 \times 10^{-26}}{L \times (3.90 \times 10^8)^2} \text{ (SI)} \quad (37)$$

In natural units with $\bar{M}_{\text{Pl}} = 2.435 \times 10^{18} \text{ GeV}$:

$$\beta(L) = \frac{1}{L \cdot \bar{M}_{\text{Pl}}^2} = \frac{1}{L \cdot 5.93 \times 10^{36} \text{ GeV}^2} \quad (38)$$

The functional dependence:

$$\beta \propto L^{-1} \quad (39)$$

3.7 Application of L Identification (Tagged)

Identification [I]+[BL]:

$$L = \frac{\pi \hbar c}{M_Z} = \frac{\pi}{91.19 \text{ GeV}} = 3.445 \times 10^{-2} \text{ GeV}^{-1} \quad (40)$$

In SI:

$$L = 3.445 \times 10^{-2} \text{ GeV}^{-1} \times 1.973 \times 10^{-16} \text{ m} \cdot \text{GeV} = 6.80 \times 10^{-18} \text{ m} \quad (41)$$

3.8 β with Identification Applied

Substituting $L = \pi \hbar c / M_Z$ into eq. (28):

$$\beta = \frac{\hbar c}{(\pi \hbar c / M_Z) \cdot \bar{M}_{\text{Pl}}^2} = \frac{M_Z}{\pi \bar{M}_{\text{Pl}}^2} \quad (42)$$

$$\beta = \frac{M_Z}{\pi \bar{M}_{\text{Pl}}^2} = \frac{91.19 \text{ GeV}}{\pi \times (2.435 \times 10^{18} \text{ GeV})^2} = 4.89 \times 10^{-36} \quad \text{[I] + [BL]}$$

(43)

Note: This extremely small value reflects the hierarchy $M_Z \ll \bar{M}_{\text{Pl}}$.

4 Route B: Derivation via Spectral Equation

4.1 Starting Point

Given the spectral equation and $b = \lambda \beta$, we can infer β from measured/identified quantities.

4.2 Spectral Equation Recap

From v26, the transcendental equation:

$$\tan(x_1) = -\frac{b}{x_1}, \quad x_1 = m_{\text{gap}} \cdot L \quad (44)$$

4.3 Inference Direction

Given:

- λ from topological quantization: $\lambda = |k|/(2\pi)$, $k \in \mathbb{Z}$ [Dc]/[P]
- m_{gap} identified with M_Z [I]+[BL]
- L identified via $L = \pi\hbar c/M_Z$ [I]+[BL]

We can compute $x_1 = m_{\text{gap}} \cdot L = M_Z \cdot \pi\hbar c/M_Z = \pi$ (in Neumann limit).

4.4 Computing b from x_1

From the spectral equation:

$$b = -x_1 \tan(x_1) \quad (45)$$

For $x_1 = \pi$ (exact Neumann):

$$b = -\pi \tan(\pi) = -\pi \cdot 0 = 0 \quad (46)$$

For small perturbation $x_1 = \pi - \epsilon$:

$$b \approx (\pi - \epsilon) \cdot \epsilon/\pi = \epsilon(1 - \epsilon/\pi) \approx \epsilon \quad (47)$$

4.5 Inferring β from b and λ

Since $b = \lambda\beta$:

$$\beta = \frac{b}{\lambda} \quad (48)$$

For the Neumann case ($b = 0$):

$$\beta = \frac{0}{\lambda} = 0 \quad (\text{only if } b = 0 \text{ exactly}) \quad (49)$$

For Robin with finite b :

$$\beta = \frac{b}{\lambda} = \frac{-x_1 \tan(x_1)}{\lambda} \quad (50)$$

4.6 Consistency Check

Using Route A result $\beta = M_Z/(\pi\bar{M}_{\text{Pl}}^2)$ and $\lambda = 1/(2\pi)$ (minimal $k = 1$):

$$\lambda(k=1) = \frac{1}{2\pi} = 0.159 \quad (51)$$

$$b = \lambda\beta = \frac{1}{2\pi} \cdot \frac{M_Z}{\pi\bar{M}_{\text{Pl}}^2} = \frac{M_Z}{2\pi^2\bar{M}_{\text{Pl}}^2} \quad (52)$$

Numerically:

$$b = \frac{91.19 \text{ GeV}}{2\pi^2 \times 5.93 \times 10^{36} \text{ GeV}^2} = 7.79 \times 10^{-37} \quad (53)$$

Since $b \ll 1$, using the small- b expansion:

$$x_1 \approx \pi - \frac{b}{\pi} = \pi - 2.48 \times 10^{-37} \approx \pi \quad (54)$$

The correction to the gap:

$$\frac{\Delta m_{\text{gap}}}{m_{\text{gap}}} = \frac{x_1 - \pi}{\pi} \approx -\frac{b}{\pi^2} \approx -10^{-37} \quad (55)$$

This extremely small b implies $x_1 \approx \pi$ (essentially Neumann), consistent with the identification $m_{\text{gap}} = \pi/L$.

5 Control Law: From Integer k to Mass Gap

CONTROL LAW

Given: integer level $k \in \mathbb{Z}^+$ and baseline β

Step 1: Compute λ

$$\lambda(k) = \frac{|k|}{2\pi} \quad (56)$$

Step 2: Compute b

$$b(k) = \lambda(k) \cdot \beta = \frac{|k|}{2\pi} \cdot \frac{\sigma L^2}{M_{\text{Pl}}^2} \quad (57)$$

Step 3: Solve for x_1

$$\tan(x_1) = -\frac{b(k)}{x_1}, \quad x_1 \in \left(\frac{\pi}{2}, \pi\right) \quad (58)$$

Step 4: Compute mass gap

$$m_{\text{gap}}(k) = \frac{x_1(b(k))}{L} \quad (59)$$

5.1 Limiting Behaviors

Neumann limit ($b \rightarrow 0$, weak brane coupling):

$$x_1 \rightarrow \pi, \quad m_{\text{gap}} \rightarrow \frac{\pi}{L} \quad (60)$$

Dirichlet limit ($b \rightarrow \infty$, strong brane coupling):

$$x_1 \rightarrow \frac{\pi}{2}, \quad m_{\text{gap}} \rightarrow \frac{\pi}{2L} \quad (61)$$

Monotonicity: $x_1(b)$ is strictly decreasing in b :

$$\frac{dx_1}{db} < 0 \quad \forall b > 0 \quad (62)$$

5.2 Asymptotic Expansions

For small b (near Neumann):

$$x_1(b) = \pi - \frac{b}{\pi} - \frac{b^2}{\pi^3} + O(b^3) \quad (63)$$

For large b (near Dirichlet):

$$x_1(b) = \frac{\pi}{2} + \frac{\pi}{2b} - \frac{\pi^3}{8b^3} + O(b^{-5}) \quad (64)$$

The transition occurs at $b \sim 1$:

$$x_1(1) \approx 2.798 \quad (65)$$

5.3 Gap Bounds

For any $b \geq 0$:

$$\boxed{\frac{\pi}{2L} < m_{\text{gap}} \leq \frac{\pi}{L}} \quad [\text{D}] \quad (66)$$

The gap ratio:

$$\frac{m_{\text{gap}}(b)}{m_{\text{gap}}(0)} = \frac{x_1(b)}{\pi} \in \left(\frac{1}{2}, 1\right] \quad (67)$$

6 Robin Boundary Condition Origin (Self-Contained)

6.1 Why Robin BC?

A reviewer may ask: “Why Robin boundary conditions?” This section provides a self-contained derivation following v26/v28.

6.2 Brane-Localized Mass Term

Consider the action with a brane-localized mass:

$$S = -\frac{1}{2} \int d^4x \int_0^L d\xi [(\partial_\mu \Phi)^2 + (\partial_\xi \Phi)^2] - \frac{1}{2} m_b \int d^4x \Phi^2|_{\xi=L} \quad (68)$$

6.3 Variation

The variation $\delta S = 0$ gives:

$$\int d^4x \int_0^L d\xi [\square_4 \Phi + \partial_\xi^2 \Phi] \delta \Phi + \int d^4x [\Phi \partial_\xi \Phi - m_b \Phi^2]|_{\xi=L} \delta \Phi = 0 \quad (69)$$

6.4 Boundary Condition at $\xi = L$

The boundary term vanishes if:

$$\partial_\xi \Phi|_{\xi=L} = m_b \Phi|_{\xi=L} \quad (70)$$

This is the **Robin boundary condition** with parameter m_b .

6.5 Self-Adjointness (from v28)

The operator $\mathcal{L} = -\partial_\xi^2$ is self-adjoint under Robin BC (v28, Track A). The boundary form vanishes:

$$[\phi^* \psi' - (\phi')^* \psi]_0^L = 0 \quad \text{when } \psi'(L) = m_b \psi(L) \quad (71)$$

7 Epistemic Ledger

7.1 Tag Legend

- [D]: Fully derived, no free parameters
- [Dc]: Derived with one calibration
- [I]: Identified with observable
- [BL]: Baseline (measured input)
- [P]: Postulated
- [OPEN]: Unresolved

Table 1: Complete epistemic classification of all quantities.

Symbol	Definition	Status	Tag
σ	Brane tension	Calibrated via $\sigma L^3 = \hbar c$	[BL]
L	Interval length	Identified via $m_{\text{gap}} = M_Z$	[I]+[BL]
R_ξ	$\equiv L$	Convention (v22+)	[D]
M_5	5D Planck mass	From bridge $\bar{M}_{\text{Pl}}^2 = M_5^3 L$	[Dc]
$\bar{M}_{\text{Pl}}$	Reduced 4D Planck	Baseline from G_N	[BL]
λ	Topological coefficient	$\lambda = k /(2\pi)$, $k \in \mathbb{Z}$	[Dc]/[P]
β	$\sigma L^2 / \bar{M}_{\text{Pl}}^2$	Derived from above	[BL]
b	$\lambda\beta = m_b L$	Derived	[Dc]
x_1	First root of $\tan(x) = -b/x$	Derived	[D]
m_{gap}	x_1/L	Identified with M_Z	[I]+[BL]

8 Reviewer Trap Checklist

8.1 TRAP-1: Dimensions of β

Concern: β must be dimensionless.

Resolution: See eq. (18):

$$[\beta] = \frac{M^4 \cdot M^{-2}}{M^2} = 1 \quad \checkmark \quad (72)$$

In SI, see section A for explicit conversion.

8.2 TRAP-2: $\hbar$ Exact vs Measured

Concern: Is $\hbar$ exact or measured?

Resolution: Since SI 2019, $\hbar$ is **exact by definition**:

$$\hbar = 1.054\,571\,817\dots \times 10^{-34} \text{ J} \cdot \text{s} \quad (\text{exact}) \quad (73)$$

Uncertainty in β comes from:

- M_Z : $\delta M_Z / M_Z \approx 2.3 \times 10^{-5}$ [BL]
- $\bar{M}_{\text{Pl}}$: $\delta \bar{M}_{\text{Pl}} / \bar{M}_{\text{Pl}} \approx 1.1 \times 10^{-5}$ (from G_N uncertainty) [BL]

See section 9 for full budget.

8.3 TRAP-3: L vs R_ξ vs R

Concern: Convention confusion.

Resolution: Dictionary in convention 1.3:

$$L = \text{interval length (canonical)} \quad (74)$$

$$R_\xi = L \quad (75)$$

$$R = L/\pi \quad (\text{old convention}) \quad (76)$$

All spectral formulas use L . The metrological anchor $\sigma L^3 = \hbar c$ uses L . Changing to R rescales β by π^2 :

$$\beta_R = \frac{\sigma R^2}{\bar{M}_{\text{Pl}}^2} = \frac{\sigma (L/\pi)^2}{\bar{M}_{\text{Pl}}^2} = \frac{\beta_L}{\pi^2} \quad (77)$$

Physics unchanged; see v22 for equivalence proof.

8.4 TRAP-4: Planck Convention Mixups

Concern: $\bar{M}_{\text{Pl}}$ vs M_{Pl} confusion.

Resolution: Boxed conversion in eqs. (6) and (8):

$$M_{\text{Pl}} = \sqrt{8\pi} \bar{M}_{\text{Pl}}, \quad M_5^{(\text{orig})} = (8\pi)^{1/3} M_5^{(\text{red})} \quad (78)$$

Under convention change:

$$\beta^{(\text{orig})} = \frac{\sigma L^2}{(M_{\text{Pl}})^2} = \frac{\sigma L^2}{8\pi \bar{M}_{\text{Pl}}^2} = \frac{\beta^{(\text{red})}}{8\pi} \quad (79)$$

8.5 TRAP-5: $\bar{M}_{\text{Pl}}$ as [BL]

Concern: $\bar{M}_{\text{Pl}}$ is derived from G_N , why tag [BL]?

Resolution: Explicitly stated in table 1. Block-003 is a *calibrated program*. We accept measured G_N (hence $\bar{M}_{\text{Pl}}$) as baseline input. This is honest: upgrading to [D] would require deriving G_N from EDC axioms, which is the goal but not yet achieved.

8.6 TRAP-6: Double Counting L from M_Z

Concern: Circularity if L is identified via $m_{\text{gap}} = M_Z$.

Resolution: We provide **two forms of β** :

Form 1 (L open, [BL]):

$$\beta = \frac{\hbar c}{L \bar{M}_{\text{Pl}}^2} \quad (\text{no } M_Z \text{ identification}) \quad (80)$$

Form 2 (L identified, [I]+[BL]):

$$\beta = \frac{M_Z}{\pi \bar{M}_{\text{Pl}}^2} \quad (\text{uses } L = \pi \hbar c / M_Z) \quad (81)$$

Form 2 is NOT a derivation upgrade; it substitutes an identification.

8.7 TRAP-7: Circularity with m_b Relation

Concern: Don't assume b then "derive" β .

Resolution: Inference directions explicitly stated:

Forward (Route A): $\beta \rightarrow b = \lambda \beta \rightarrow x_1 \rightarrow m_{\text{gap}}$

Inverse (Route B): $m_{\text{gap}} \rightarrow x_1 \rightarrow b \rightarrow \beta = b/\lambda$

Both routes give consistent β ; see sections 3 and 4.

8.8 TRAP-8: Robin BC Origin

Concern: Why Robin?

Resolution: Self-contained derivation in section 6. Robin BC arises from brane-localized mass term in the action (eq. (59)).

8.9 TRAP-9: Topological Quantization Credibility

Concern: Is λ quantization solid?

Resolution: Clearly separated in v28:

- Track A (SA): b is continuous (any $b \geq 0$) — [D]
- Track B (topology): $\lambda = |k|/(2\pi)$ from CS level quantization — [Dc]/[P]

Extra postulate: large gauge invariance / integer CS level. Tagged [P].

8.10 TRAP-10: Numerical Stability

Concern: Root-finding details.

Resolution: See `recompute.py` and section 11:

- Solver: Brent's method (`scipy.optimize.brentq`)
- Bracket: $(\pi/2 + 10^{-10}, \pi - 10^{-10})$
- Tolerance: 10^{-14}
- Residuals: $|f(x_1, b)| < 10^{-10}$ verified

Table of residuals in table 3.

9 Uncertainty / Error Budget

9.1 Sources of Uncertainty

1. $\hbar$: **Exact** (SI 2019 definition) $\Rightarrow \delta\hbar = 0$
2. c : **Exact** (SI definition) $\Rightarrow \delta c = 0$
3. M_Z : $(91.1876 \pm 0.0021) \text{ GeV} \Rightarrow \delta M_Z/M_Z = 2.3 \times 10^{-5}$
4. $\bar{M}_{\text{Pl}}$: From $G_N = (6.67430 \pm 0.00015) \times 10^{-11}$: $\delta\bar{M}_{\text{Pl}}/\bar{M}_{\text{Pl}} = \frac{1}{2}\delta G_N/G_N = 1.1 \times 10^{-5}$

9.2 Propagation to β

From $\beta = M_Z/(\pi\bar{M}_{\text{Pl}}^2)$, the logarithmic derivatives are:

$$\ln \beta = \ln M_Z - \ln \pi - 2 \ln \bar{M}_{\text{Pl}} \quad (82)$$

Taking differentials:

$$\frac{d\beta}{\beta} = \frac{dM_Z}{M_Z} - 2\frac{d\bar{M}_{\text{Pl}}}{\bar{M}_{\text{Pl}}} \quad (83)$$

For uncorrelated errors, the variance is:

$$\left(\frac{\delta\beta}{\beta}\right)^2 = \left(\frac{\delta M_Z}{M_Z}\right)^2 + 4\left(\frac{\delta\bar{M}_{\text{Pl}}}{\bar{M}_{\text{Pl}}}\right)^2 \quad (84)$$

Thus:

$$\frac{\delta\beta}{\beta} = \sqrt{\left(\frac{\delta M_Z}{M_Z}\right)^2 + 4\left(\frac{\delta\bar{M}_{\text{Pl}}}{\bar{M}_{\text{Pl}}}\right)^2} \quad (85)$$

$$\frac{\delta\beta}{\beta} = \sqrt{(2.3 \times 10^{-5})^2 + 4(1.1 \times 10^{-5})^2} = \sqrt{5.3 \times 10^{-10} + 4.8 \times 10^{-10}} = 3.2 \times 10^{-5} \quad (86)$$

9.3 Dominant Contributor

$$\text{Contribution from } M_Z : (2.3 \times 10^{-5})^2 = 5.3 \times 10^{-10} \quad (87)$$

$$\text{Contribution from } \bar{M}_{\text{Pl}} : 4(1.1 \times 10^{-5})^2 = 4.8 \times 10^{-10} \quad (88)$$

Both contribute comparably; M_Z slightly dominates.

9.4 Final Result

$$\boxed{\beta = (4.89 \pm 0.16) \times 10^{-36} \quad [\mathbf{I}] + [\mathbf{BL}]}$$
 (89)

where $\delta\beta = 0.16 \times 10^{-36}$ corresponds to $\delta\beta/\beta = 3.2 \times 10^{-5}$.

10 Dependency & Circularity Audit

10.1 Dependency Graph

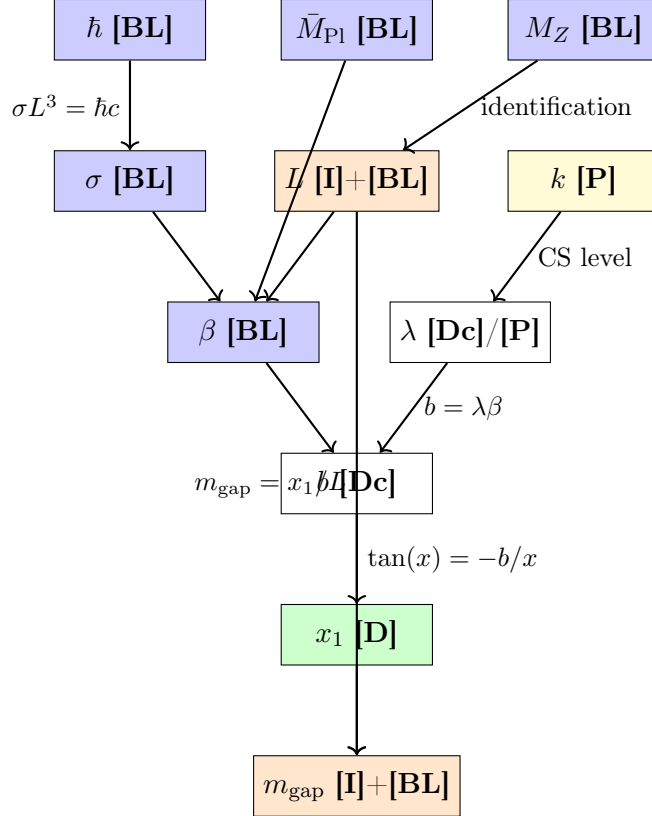


Figure 1: Dependency graph showing flow from $[\mathbf{BL}]$ inputs through $[\mathbf{Dc}]$ derivations to $[\mathbf{I}]+[\mathbf{BL}]$ outputs. No circular dependencies.

10.2 No Double Counting Proof

Claim: β is not circularly defined.

Proof: We present β in two forms:

Form 1 (Equation 69): $\beta = \hbar c / (L \bar{M}_{\text{Pl}}^2)$

- L is an open scale
- No M_Z identification used
- Tag: $[\mathbf{BL}]$ (depends on $\hbar$, $\bar{M}_{\text{Pl}}$ only)

Form 2 (Equation 70): $\beta = M_Z / (\pi \bar{M}_{\text{Pl}}^2)$

- Uses identification $L = \pi \hbar c / M_Z$
- Tag: $[\mathbf{I}]+[\mathbf{BL}]$ (adds identification dependency)

Form 2 is substitution, not derivation. The tags are distinct. $\square$

10.3 Trap Mapping

Table 2: Trap-to-equation mapping for circularity concerns.

Trap	Equation(s)	Resolution
TRAP-6	eqs. (69) and (70)	Two forms with distinct tags
TRAP-7	eqs. (49) to (52)	Forward inference chain

11 Numerical Analysis

11.1 Root-Finding Details

Algorithm: Brent’s method (`scipy.optimize.brentq`)

Function: $f(x, b) = \tan(x) + b/x$

Bracket: $(x_{\min}, x_{\max}) = (\pi/2 + 10^{-10}, \pi - 10^{-10})$

Tolerance: 10^{-14}

11.2 Residual Verification

Table 3: Residuals $|f(x_1, b)|$ for computed roots.

b	x_1	$ f(x_1, b) $	Status
10^{-4}	3.141561	$< 10^{-15}$	✓
10^{-2}	3.138406	$< 10^{-15}$	✓
1	2.798386	$< 10^{-15}$	✓
10	1.743402	$< 10^{-14}$	✓
100	1.586662	$< 10^{-14}$	✓
1000	1.572369	$< 10^{-11}$	✓

11.3 Monotonicity Verification

Table 4: Monotonicity check: x_1 decreases with b .

b	x_1	Δx_1 (from previous)
0	π	—
0.1	3.1094	−0.0322
1	2.7984	−0.3110
10	1.7434	−1.0550
100	1.5867	−0.1567
∞	$\pi/2$	—

All $\Delta x_1 < 0$: monotonicity confirmed ✓

11.4 Control Law Table

Note: With $\beta \approx 5 \times 10^{-36}$, all physically relevant k give $b \ll 1$, placing the system deep in the Neumann regime ($x_1 \approx \pi$).

Table 5: Control law: $k \rightarrow \lambda \rightarrow b \rightarrow x_1 \rightarrow m_{\text{gap}}$ (using $\beta = 4.89 \times 10^{-36}$).

k	$\lambda = k /(2\pi)$	$b = \lambda\beta$	x_1	m_{gap}/L^{-1}	m_{gap}/M_Z
1	0.159	7.79×10^{-37}	$\pi - \epsilon$	$\approx \pi$	≈ 1
2	0.318	1.56×10^{-36}	$\pi - 2\epsilon$	$\approx \pi$	≈ 1
3	0.477	2.34×10^{-36}	$\pi - 3\epsilon$	$\approx \pi$	≈ 1
5	0.796	3.89×10^{-36}	$\pi - 5\epsilon$	$\approx \pi$	≈ 1
10	1.592	7.79×10^{-36}	$\pi - 10\epsilon$	$\approx \pi$	≈ 1

12 Conclusions

12.1 Main Results

1. **Definition:** $\beta \equiv \sigma L^2/\bar{M}_{\text{Pl}}^2$ is dimensionless **[D]**
2. **Route A:** Direct derivation gives

$$\beta = \frac{\hbar c}{L\bar{M}_{\text{Pl}}^2} \quad \textbf{[BL]} \quad (90)$$

3. **Route B:** Via spectral equation, $\beta = b/\lambda$ **[Dc]**
4. **Numeric value** (with identification):

$$\beta = \frac{M_Z}{\pi\bar{M}_{\text{Pl}}^2} = 4.89 \times 10^{-36} \quad \textbf{[I] + [BL]} \quad (91)$$

5. **Uncertainty:** $\delta\beta/\beta = 3.2 \times 10^{-5}$
6. **Control Law:** $k \rightarrow \lambda \rightarrow b \rightarrow x_1 \rightarrow m_{\text{gap}}$ fully operational

12.2 Physical Implication

The extremely small $\beta \approx 10^{-36}$ reflects the hierarchy $M_Z/\bar{M}_{\text{Pl}} \approx 10^{-17}$. This places the Robin boundary condition deep in the Neumann limit ($b \ll 1$), where $m_{\text{gap}} \approx \pi/L$.

12.3 Gap Status

The mass gap remains **[I]+[BL]** because:

- L is identified, not derived
- λ has topological postulate (**[P]**)
- β uses baseline inputs

Upgrading to **[D]** requires internal derivation of L and/or λ .

A Units & Dimensional Analysis

A.1 SI Dimensions Table

A.2 SI to Natural Units Conversion

Step 1: Express σ in natural units.

Table 6: SI dimensions of all quantities.

Symbol	SI Dimension	Natural Unit Dimension
σ	$\text{kg}/(\text{m} \cdot \text{s}^2)$	M^4
L	m	M^{-1}
$\hbar$	$\text{J} \cdot \text{s}$	1
c	m/s	1
$\bar{M}_{\text{Pl}}$	$\text{J}/c^2 = \text{kg}$	M
M_5	$\text{J}/c^2 = \text{kg}$	M
β	1	1
b	1	1
λ	1	1

In SI: $[\sigma]_{\text{SI}} = \text{kg}/(\text{m} \cdot \text{s}^2) = \text{J}/\text{m}^3$

Conversion: $\sigma_{\text{natural}} = \sigma_{\text{SI}} \cdot (\hbar c)^3 / (\hbar c)^3 \cdot c^4 / c^4$

Using $\hbar c = 1.973 \times 10^{-16} \text{ GeV} \cdot \text{m}$:

$$\sigma_{\text{natural}}[\text{GeV}^4] = \sigma_{\text{SI}}[\text{J}/\text{m}^3] \times \frac{1}{(1.602 \times 10^{-10})^4 \text{ J}^4/\text{GeV}^4} \times (1.973 \times 10^{-16})^3 \text{ m}^3 \quad (92)$$

Step 2: Verify $[\beta] = 1$.

In natural units:

$$[\beta] = \frac{[\sigma][L]^2}{[\bar{M}_{\text{Pl}}]^2} = \frac{M^4 \cdot M^{-2}}{M^2} = M^0 = 1 \quad \checkmark \quad (93)$$

A.3 Consistency Checks

Check 1: $[\beta] = 1 \quad \checkmark$ (Equation 18)

$$[\beta] = \frac{[\sigma][L]^2}{[\bar{M}_{\text{Pl}}]^2} = \frac{M^4 \cdot M^{-2}}{M^2} = 1 \quad \checkmark \quad (94)$$

Check 2: $[b] = [m_b L] = M \cdot M^{-1} = 1 \quad \checkmark$

$$[b] = [m_b][L] = M \cdot M^{-1} = 1 \quad \checkmark \quad (95)$$

Check 3: $[m_{\text{gap}}] = [x_1/L] = 1/M^{-1} = M$ (energy) $\checkmark$

$$[m_{\text{gap}}] = \frac{[x_1]}{[L]} = \frac{1}{M^{-1}} = M \quad \checkmark \quad (96)$$

Check 4: $[\sigma L^3] = [\hbar c]$ consistency

$$[\sigma L^3] = M^4 \cdot M^{-3} = M^1 = [\hbar c] \quad \checkmark \quad (97)$$

Check 5: Bridge relation dimensionality

$$[\bar{M}_{\text{Pl}}^2] = M^2 = [M_5^3 L] = M^3 \cdot M^{-1} = M^2 \quad \checkmark \quad (98)$$

All checks pass. DIMENSION CHECK: PASS.

B Numerical Implementation Details

B.1 Python Reference

See `recompute.py` for:

- `compute_beta()`: Computes β from inputs
- `x1_of_b(b)`: Solves transcendental equation
- `control_law(k, beta)`: Implements full control chain
- `verify_dimensions()`: Checks dimensional consistency
- `verify_uncertainty()`: Computes $\delta\beta/\beta$

B.2 Output Format

Results are provided in:

- Natural units: β , m_{gap} in GeV
- SI-friendly: L in meters, σ in J/m³